Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

A. Epstein-Barr virus (EBV)

B. Measles virus

C. Human immunodeficiency virus (HIV)

D. Human papillomavirus

Reasoning + Answer:

<think>

- CD21 is a receptor for the complement component C3d, which is involved in signaling the B cell to undergo class-switching to produce IgG antibodies.
- EBV and measles are both viruses that can bind to CD21.
- EBV specifically uses CD21 as a receptor for entry into B cells.
- HIV uses CD4 and CCR5 or CXCR4, while human papillomavirus does not use CD21 as a receptor.

- Given this information, the genetic modification of B cells to not express CD21 would most affect the pathogenesis of EBV.
- Therefore, the answer is A.

</think><answer>A</answer>